



Hands-On Lab Exercise 3: Understanding Variables in Shell Script

Topic: Working with Variables in Shell Script

Author: Dr. Sandeep Kumar Sharma

This lab introduces the concept of **variables in shell scripting** using simple explanations and practical examples.

Variables are one of the most important building blocks of automation. They allow scripts to **store data** and **reuse values** while executing commands.



What is a Variable?

A **variable** is a name that stores a value.

In simple words:

A variable is a container that holds some data.

Example idea: If you store a name, path, number, or message — that is a variable.



Why Do We Need Variables?

We need variables because they: - Store information - Reuse values multiple times - Make scripts flexible - Reduce hard-coded values - Make scripts easy to change - Improve automation logic

Without variables, scripts become: - Fixed - Rigid - Hard to manage - Hard to scale



Purpose of Using Variables

Variables help to: - Store user names - Store paths - Store file names - Store messages - Store commands output - Store system information

They make scripts **dynamic**, not static.

Understanding Variable Concept

Think like this:

"Instead of writing the same value again and again, store it once and reuse it."

That is the purpose of variables.



Learning Objective

- Understand what a variable is
 - Understand why variables are needed
 - Learn how to declare variables
 - Learn how to define variables
 - Learn how to use variables
 - Learn variable usage in scripts
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Learning Outcome

After this lab, participants will: - Create variables in shell script - Store values in variables - Use variables in commands - Build flexible scripts - Write clean automation scripts



Variable Rules in Shell Script

1. No space around `=`
2. Variable names should not start with numbers
3. No special characters in variable names
4. Use meaningful names
5. Variables are case-sensitive

Correct:

```
name="Linux"  
PATH1="/home/user"
```

Wrong:

```
name = "Linux"  
1name="test"  
my-name="abc"
```



Example 1: Simple Variable Declaration

Step 1: Create Script

```
touch var1.sh
```

Step 2: Open File

```
nano var1.sh
```

Step 3: Write Script

```
#!/bin/bash  
  
message="Welcome to Shell Scripting"  
  
# printing variable value  
echo "$message"
```

Step 4: Permission

```
chmod +x var1.sh
```

Step 5: Run

```
./var1.sh
```

Explanation:

- `message` is a variable name
 - `=` assigns value
 - `"Welcome to Shell Scripting"` is the value
 - `$message` is used to access variable value
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Example 2: Multiple Variables

Step 1: Create Script

```
touch var2.sh
```

Step 2: Open File

```
nano var2.sh
```

Step 3: Write Script

```
#!/bin/bash

name="Linux"
trainer="Dr. Sandeep Kumar Sharma"
course="Shell Scripting"

# using variables
echo "Course Name: $course"
echo "Trainer: $trainer"
echo "Platform: $name"
```

Step 4: Permission

```
chmod +x var2.sh
```

Step 5: Run

```
./var2.sh
```



Example 3: Variables in System Tasks

Step 1: Create Script

```
touch var3.sh
```

Step 2: Open File

```
nano var3.sh
```

Step 3: Write Script

```
#!/bin/bash

folder="project"
sub1="app"
sub2="logs"
sub3="config"

mkdir $folder
mkdir $folder/$sub1
mkdir $folder/$sub2
mkdir $folder/$sub3

echo "Project structure created using variables"
```

Step 4: Permission

```
chmod +x var3.sh
```

Step 5: Run

```
./var3.sh
```

Explanation:

Variables store directory names and reuse them

Instead of hardcoding:

```
mkdir project/app  
mkdir project/logs
```

We use variables for flexibility.

Key Understanding

- Variables store values
 - `$` is used to access variable
 - Variables make scripts reusable
 - Variables make scripts dynamic
 - Variables are core to automation
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Summary

- Variable = data container
 - Variables improve automation
 - Variables reduce repetition
 - Variables make scripts flexible
 - Variables are automation foundation
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End of Lab

Variables are the **next step in automation journey**.

After variables, automation becomes powerful.

Next labs will include: - User input - Conditions - Loops - Functions - Real automation systems

Author:

Dr. Sandeep Kumar Sharma



LinkedIn:

linkedin.com/in/sandeep-kaushik-2a345856

Medium Blog:

medium.com/@shyamsandeep28